

The Erdős #249/#257 Lean formalization:

what is proved, what is cited, and what stays open

ai_workflow (upstream of [Plectis](#))

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Companion to the Lean 4 repository [plectis-lean-erdos249-257](#), pinned at commit [de7a360](#).

Abstract

This note describes, in ordinary mathematical language, a machine-checked Lean 4 development around two Erdős irrationality problems: the irrationality of Erdős–Borwein-type series $\sum 1/(b^{a_k} - 1)$ (Erdős #257), and the still-open question of whether the totient constant $S = \sum_{n \geq 1} \varphi(n)/2^n$ is irrational (Erdős #249). The development proves the Erdős–Borwein constant irrational for every integer base $b \geq 2$, and the same for several named infinite supports. It rewrites S as a weighted Erdős–Borwein object; proves *unconditionally* that S equals no rational with denominator up to about 7.96×10^{34} ; and *reduces* the irrationality of S to a sequence of finite, decidable certificates. The remaining analytic question is whether that certificate supply is unbounded. Neither open problem is solved. Machine-checked non-claims in the repository state the same boundary. Every formal statement links to the exact Lean source line. The Lean kernel verifies the package with no `sorry`, no `admit`, and no `native_decide`.

Contents

1	Main results and non-results	2
2	How to read this document	3
3	Two constants, two problems	3
4	The Mersenne–Lambert ladder	4
5	Erdős–Borwein-type irrationality (the #257 direction)	5
5.1	Full support: the Erdős–Borwein constant, every base	5
5.2	Named infinite-support cases	5
5.3	What stays open	6
6	The totient constant S (Erdős #249)	6
6.1	Unconditional: no small denominator	6
6.2	The geometric picture: S as coprime-pair mass	6
6.3	Conditional: reduction to finite certificates	7
7	Rationality as a carry system (further infrastructure)	8
7.1	Tail-orbit rigidity: a rationality criterion	9
7.2	Boolean–Möbius carry coordinates (the #257 direction)	9
7.3	Residue wraps, reciprocal mass, and unbounded carries	10
7.4	Greedy geometry of the #257 value set	12
8	Where this sits: the Erdős landscape	13
9	What is proved, cited, and open	13

10 Verification and reproducibility	14
11 A sibling project, in progress	14
A Index of principal declarations	14

1 Main results and non-results

This section states what the Lean development proves and what it does not. Every item is machine-checked at the pinned commit; later sections expand each one and link it to source. The two named problems are Erdős #249 (is $S = \sum_{n \geq 1} \varphi(n)/2^n$ irrational?) and Erdős #257 (is $\sum_{n \in A} 1/(2^n - 1)$ irrational for *every* infinite $A \subseteq \mathbb{N}$?). Both are open, and this development solves neither.

The Mersenne–Lambert ladder. $L(f) = \sum_{n \geq 1} \frac{f(n)}{2^n - 1}$			
$L(\mu) = \frac{1}{2}$	$L(\varphi) = 2$	$L(1) = E$	$L(\varphi * \mu) = S$ (#249)
rational	rational	irrational (proved)	open
The #249 reduction chain.			
$S \in \mathbb{Q} \implies \text{eventual binary periodicity} \implies R_{N+h} - R_N \in \mathbb{Z}$			
$\implies \text{a finite certifiedKill witnesses failure} \iff R_{N+h} - R_N \notin \mathbb{Z}$			
(completeness: the finite certificate is exact, not a truncation heuristic; #249 becomes “the certificate supply is unbounded”)			

Proved (machine-checked).

- **All-base Erdős–Borwein irrationality.** $\sum_{n \geq 1} 1/(b^n - 1)$ is irrational for every integer $b \geq 2$; the Erdős–Borwein constant E is the base-2 case (Result 4). This is a formalization of Erdős’s classical theorem, and it is exactly the integer-base slice of Erdős #1049.
- **A #257 support calculus.** $\sum_{n \in A} 1/(b^n - 1)$ is irrational, at every base $b \geq 2$, for several structured infinite supports A : factorial, powers of two, multiples, pairwise-coprime with summable reciprocals, eventually periodic, residue classes, and the odd numbers (Result 5). These are named cases, not the universal #257 theorem.
- **A change of coordinates for #249.** The totient constant is a weighted Erdős–Borwein object, $S = L(\varphi * \mu)$ with a nonnegative primitive-conductor weight, and equals $\frac{1}{2} + \sum_d \mu(d)/(2^d - 1)^2$ and $\frac{1}{2} + \Pr(\gcd(X, Y) = 1)$ for independent fair-coin waiting times (Results 2, 3, 7).
- **An unconditional denominator exclusion.** S equals no rational of denominator up to about 7.96×10^{34} (Result 6). This is a record on the denominator, not a proof of irrationality.
- **A reduction of #249 to a finite, decidable supply.** Rationality of S forces an integer tail-period law; a finite certificate witnesses its failure, and by a completeness theorem the certificate exists *exactly* when the tail difference is non-integral (Results 8–11). The initial segment of the required supply is machine-checked non-empty (Result 12).
- **Carry coordinates for rational values.** A rationality criterion for binary series with linearly bounded coefficients (tempered integer carry orbits); a support-free Boolean–Möbius carry certificate exactly equivalent to a rational #257 support value; order-wrap, reciprocal-mass, and unbounded-carry constraints that any such rational value must satisfy; and a

greedy geometry for the #257 value set (Results 13–21, Section 7). These are equivalences and constraints, not exclusions of any open value.

Not proved, and not claimed. The irrationality of S (#249) and the universal #257 statement are open. The reduction turns #249 into the claim that the certificate supply is *unbounded*. That unboundedness is the remaining analytic question, and it is left open. The repository records these boundaries as machine-checked non-claims (`not_erdos_249_solution`, `not_erdos_257_solution`; [NON_CLAIMS.md](#)).

How it is checked. The Lean 4 kernel verifies the package against Mathlib with no `sorry`, no `admit`, and no `native_decide`; every finite computation is discharged by `decide`. Section 10 gives the reproduction commands.

2 How to read this document

This note is written so that a number theorist who has never used a proof assistant can see what has and has not been proved, and can check any claim without reading Lean. Two conventions make that possible.

First, the mathematics is stated in ordinary notation. Lean names appear only as pointers. Second, every formalized statement carries a link of the form [CertificateKernel.lean:8006](#) (`irrational_erdosBorwein_series`). The blue `file:line` text links to that line of source on GitHub, pinned to commit [de7a360](#); the name in parentheses is the Lean declaration. Line numbers refer to that commit.

The boundary between what is proved, what is only cited, and what is open is stated carefully. The repository enforces it in code through non-claims ([NON_CLAIMS.md](#)); Section 9 repeats it here.

3 Two constants, two problems

Work at base 2 unless a base b is named. Two Dirichlet-type series are central.

- The **Erdős–Borwein constant** $E = \sum_{n \geq 1} \frac{1}{2^n - 1}$.
- The **totient constant** $S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$, where φ is Euler’s totient.

The #257 direction. Erdős #257 asks whether $\sum_{n \in A} 1/(2^n - 1)$ is irrational for *every* infinite $A \subseteq \mathbb{N}$. The development proves several cases of the analogous statement $\sum_k 1/(b^{a_k} - 1)$, at every base $b \geq 2$, but not the universal statement.

The #249 direction. Whether S is irrational is open. The development gives an unconditional exclusion of small-denominator rationals and a conditional reduction of the problem to finite certificates. Neither is a solution.

The two constants sit one convolution apart on a single ladder.

4 The Mersenne–Lambert ladder

Define the Lambert-type transform of an arithmetic function f by

$$L(f) = \sum_{n \geq 1} \frac{f(n)}{2^n - 1} = \sum_{m \geq 1} \frac{(f * \mathbf{1})(m)}{2^m}, \quad (f * \mathbf{1})(m) = \sum_{d|m} f(d),$$

the second equality being the standard Lambert rearrangement (one Dirichlet convolution by the constant $\mathbf{1} = \zeta$). Walking the ladder by convolution gives five values with rational, irrational, open, or transcendental status.

input f	$L(f)$	value	status in the repository
μ (Möbius)	$L(\mu)$	$1/2$	rational, proved
φ (totient)	$L(\varphi)$	2	rational, proved
$\mathbf{1}$ (constant one)	$L(\mathbf{1})$	E	irrational, machine-checked
$A = \varphi * \mu$	$L(A)$	S	open (Erdős #249)
Id	$L(\text{Id})$	$\sum \sigma(m)/2^m$	transcendental (Nesterenko 1996 [4], cited only)

The weight $A = \varphi * \mu$ is the primitive-conductor count: $A(p) = p - 2$ on primes, $A \geq 0$, and $A * \zeta = \varphi$. One convolution by ζ separates the open constant S from the rational 2 , and one more separates rational from transcendental ($\varphi * \zeta = \text{Id}$, $\text{Id} * \zeta = \sigma$). So S is the rung one convolution from the machine-checked irrational E : the same series $\sum(\cdot)/(2^d - 1)$, with the constant weight 1 replaced by the nonnegative weight A .

Result 1 (the rational rungs). $\sum_{d \geq 1} \mu(d)/(2^d - 1) = 1/2$ and $\sum_{d \geq 1} \varphi(d)/(2^d - 1) = 2$, *exactly*.

Formalized: [MersenneLambertLadder.lean:527](#) (`tsum_moebius_div_two_pow_sub_one_eq_half`) and [MersenneLambertLadder.lean:515](#) (`tsum_totient_div_two_pow_sub_one_eq_two`); restated on the #249 series shape at [tsum_moebius_div_two_pow_sub_one_eq_half](#) and [tsum_totient_div_two_pow_sub_one_eq_two](#).

Result 2 (the positive lift, $L(A) = S$). *With the primitive-conductor weight $A = \varphi * \mu$,*

$$\sum_{d \geq 1} \frac{A(d)}{2^d - 1} = \sum_{n \geq 1} \frac{\varphi(n)}{2^n} = S.$$

Formalized: [CertificateKernel.lean:18116](#) (`tsum_primWeight_div_two_pow_sub_one_eq_totient_series`), with the weight A defined at [primWeight](#). This places S in the same $\sum(\cdot)/(2^d - 1)$ family as E , with a nonnegative weight.

Result 3 (the Möbius-square lens).

$$S = \frac{1}{2} + \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2}.$$

Formalized: [CertificateKernel.lean:18125](#) (`totient_series_eq_half_add_moebius_mersenne_square`). Writing $T = \sum_{d \geq 1} \mu(d)/(2^d - 1)^2$, irrationality of S is equivalent to irrationality of T , since $S = \frac{1}{2} + T$. The factor $1/(2^d - 1)^2$ is the probability that d divides two independent fair-coin waiting times (Section 6.2). The signed constant, the coprime-pair mass, and the tail differences of Section 6.3 are three readings of the same quantity.

5 Erdős–Borwein-type irrationality (the #257 direction)

5.1 Full support: the Erdős–Borwein constant, every base

Result 4 (full-support irrationality). *For every integer $b \geq 2$, the series $\sum_{n \geq 1} \frac{1}{b^n - 1}$ is irrational. In particular the Erdős–Borwein constant $E = \sum_{n \geq 1} 1/(2^n - 1)$ is irrational.*

Formalized: [CertificateKernel.lean:7999](#) (`irrational_erdosSum_full_support`) (all bases) and [CertificateKernel.lean:8006](#) (`irrational_erdosBorwein_series`) (the base-2 corollary). This is Erdős’s 1948 theorem [1]. In the ladder of Section 4 it is the rung $L(1)$, and it is the fixed machine-checked point one convolution away from the open S .

The proof runs through the certificate machine of the kernel rather than through a direct digit argument: a bounded Bertrand/CRT frame on a first block, a middle-window average over divisor pairs with a pigeonhole selection, and an explicit closure of the parameters. The Lambert bridge $\sum 1/(b^n - 1) = \sum_m \tau(m)/b^m$ (divisor-count series) is [erdosSum_full_support_eq_tsum_divisor_count](#) in the source.

5.2 Named infinite-support cases

Beyond full support, the development proves irrationality for a family of infinite supports A , at every base $b \geq 2$. Each is a formalized case of the #257 statement family. None is the universal statement, and none is claimed as new mathematics.

Result 5 (support-class family). *For every integer $b \geq 2$, the series $\sum_{n \in A} 1/(b^n - 1)$ is irrational for each of the following supports A :*

- (a) *factorial support $A = \{n! : n \geq 1\}$, giving $\sum 1/(b^{n!} - 1)$;*
- (b) *power-of-two support $A = \{2^n : n \geq 0\}$, giving $\sum 1/(b^{2^n} - 1)$;*
- (c) *multiples $A = d\mathbb{N}$;*
- (d) *pairwise-coprime supports with summable reciprocals (Erdős’s condition [2]);*
- (e) *eventually-periodic supports, residue classes, and the odd numbers.*

These sort into three groups. The classical case is (d): infinite pairwise-coprime A with $\sum_{a \in A} 1/a < \infty$, formalized from Erdős’s condition. The lcm-gap named cases are (a) and (b), whose support gaps outgrow the running lcm; the structured families are (c) and (e).¹ Formalized: (a) [irrational_erdosSum_factorial_support](#); (b) [irrational_erdosSum_two_pow_support](#); (c) [irrational_erdosSupportSeries_multiples](#); (d) [irrational_erdosSupportSeries_pairwise_coprime](#); (e) [irrational_erdosSupportSeries_eventuallyPeriodic](#), [irrational_erdosSupportSeries_residueClass](#), [irrational_erdosSupportSeries_odd](#). The two base-2 instances that name the problem directly are [erdos257_family_factorial_instance](#) and [erdos257_family_two_pow_instance](#).

The factorial and power-of-two cases are worth a word because they are not reachable by a Liouville shortcut: the power-of-two gap $2^k - \text{lcm}$ stays within a bounded ratio of the lcm, so the terms do not decay fast enough for a transcendence-measure argument, and a uniform criterion is needed. The mechanism is what the repository calls *period noncollapse*: a prime-valuation-deficit witness forces the multiplicative order of the base modulo the relevant modulus not to collapse, which rules out the long repeated digit blocks a rational value would require.

¹Periodic-coefficient Lambert series have their own literature; see e.g. Luca–Tachiya [6]. The cases here are formalized in the repository’s own framework and are not claimed as new results.

Remark. A source comment near [lcm_gap_hypothesis_fails_full_support](#) describes a second, near-integer route to full support whose witnesses are not built there, and says so. The full-support theorem is proved by the separate certificate engine cited under Result 4. The two coexist; the near-integer route is one lane, not a retraction of the theorem.

5.3 What stays open

The universal Erdős #257 statement, irrationality of $\sum_{n \in A} 1/(2^n - 1)$ for *every* infinite A , is not proved. The proved cases are a family, not the quantified theorem, and the repository marks this in code (the non-claim `not_erdos_257_solution` in [NON_CLAIMS.md](#)).

6 The totient constant S (Erdős #249)

Whether $S = \sum_{n \geq 1} \varphi(n)/2^n$ is irrational is open. The development gives two things and claims neither as a solution: an unconditional exclusion of small-denominator rationals, and a conditional reduction to finite certificates.

6.1 Unconditional: no small denominator

Result 6 (denominator exclusion). *For every integer p and every q with $1 \leq q \leq 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}$,*

$$S \neq \frac{p}{q}.$$

Equivalently, if S is rational then its reduced denominator exceeds that bound.

Formalized: [CertificateKernel.lean:18055](#) (`tsum_totient_div_pow_two_ne_ratCast_of_den_le_79639646646701375323355774875831053`). The bound is reached by a ladder of finite, elementary exclusions,

$$4838 \rightarrow 4\,194\,304 = 2^{22} \rightarrow 2.49 \times 10^{17} \text{ (Farey } K=120) \rightarrow 7.96 \times 10^{34} \text{ (Farey } K=240),$$

each a mediant/Farey argument that traps any candidate m/q inside a forbidden gap. See [tsu_m_totient_div_pow_two_ne_ratCast_of_den_le_4838](#) for the first rung and [gap_check_window_1_120_le_248672326362367909](#), [gap_check_window_1_240_le_79639646646701375323355774875831053](#) for the two Farey windows.

This is a record on the denominator, not a proof of irrationality. A rational with a larger denominator is not excluded by it.

Remark (one record, four faces). Because the ladder identities of Section 4 are equalities, the same finite record transfers verbatim to three other representations of the constant: to the positive lift $\sum_d A(d)/(2^d - 1)$ ([CertificateKernel.lean:18144](#)), to the signed Möbius-square constant $T = \sum_d \mu(d)/(2^d - 1)^2$ at half the bound ([CertificateKernel.lean:18158](#)), and to the coprimality probability of Section 6.2 at half the bound. Half appears because $S = \frac{1}{2} + T$ turns a denominator d for T into $2d$ for S .

6.2 The geometric picture: S as coprime-pair mass

Let X, Y be independent fair-coin waiting times, $\Pr(X = n) = 2^{-n}$ for $n \geq 1$. Then $\Pr(d \mid X) = 1/(2^d - 1)$, so the Lambert transform is a divisor calculus of X : $L(f) = \mathbb{E}[(f * \zeta)(X)]$. The rungs read as $L(\mu) = \Pr(X = 1) = 1/2$, $L(\varphi) = \mathbb{E}[X] = 2$, $L(\mathbf{1}) = \mathbb{E}[\tau(X)] = E$, and $L(A) = \mathbb{E}[\varphi(X)] = S$.

Counting visible lattice points (a, b) with $a + b = n$, $a > 0$, $\gcd(a, b) = 1$ gives exactly $\varphi(n)$, uniformly in n , so S is itself a coprime-pair mass.

Result 7 (coprimality probability).

$$S = \frac{1}{2} + \Pr(\gcd(X, Y) = 1) = \frac{1}{2} + \sum_{\substack{a, b \geq 1 \\ \gcd(a, b) = 1}} 2^{-(a+b)}.$$

Formalized: [CertificateKernel.lean:18228](#) (`totient_series_eq_half_add_visible_coprime_pairs`), with the underlying lattice identity at [tsum_visible_coprime_pairs_eq_totient_series](#). Base 2 is the one point where $\Pr(X = a) \Pr(Y = b) = 2^{-(a+b)}$ needs no normalizing constant, so the totient sum and the coprimality probability line up with no boundary correction. In this reading, Erdős #249 asks whether two independent fair-coin waiting times are coprime with irrational probability.

6.3 Conditional: reduction to finite certificates

The reduction turns irrationality of S into the non-vanishing of a decidable predicate at infinitely many parameters. Everything in it is elementary: $\varphi(n) \leq n$, geometric tails, and integer window arithmetic. There is no q-*Padé* input, no Stern–Brocot input, and no unformalized citation ([TotientTailPeriodKiller](#)).

Step 1: rationality forces an integer tail-period law. Write the fractional layer of $2^N S$ as the tail

$$R_N = \sum_{m \geq 1} \frac{\varphi(N + m)}{2^m}.$$

If S is rational, its binary expansion is eventually periodic, which forces, for a suitable period h and all large N , the integrality $R_{N+h} - R_N \in \mathbb{Z}$. To refute integrality at a finite depth L , form the window discrepancy

$$\Delta_{h,N,L} = \sum_{j=0}^{L-1} (\varphi(N + h + 1 + j) - \varphi(N + 1 + j)) 2^{L-1-j} \in \mathbb{Z},$$

the first L binary digits of $R_{N+h} - R_N$ up to a controlled deep tail (bounded using only $\varphi(n) \leq n$), and call the pair *certified* when the residue of $\Delta_{h,N,L}$ modulo 2^L misses a neighbourhood of 0:

$$\text{certifiedKill}(h, N, L) : N + h + L + 2 < (\Delta_{h,N,L} \bmod 2^L) < 2^L - (N + h + L + 2).$$

This is a finite integer computation, decidable by the kernel. It is the natural finite object here because a rational S would pin the tail difference to an integer, and a residue that stays a fixed distance from 0 is a finite proof that it did not.

Result 8 (tail-period reduction). *If for every period $h \geq 1$ and every threshold N_0 there exist $N \geq N_0$ and L with $\text{certifiedKill}(h, N, L)$, then S is irrational.*

Formalized: [TotientTailPeriodKiller.lean:386](#) (`irrational_totient_series_of_certificate_supply`); the tail R_N and the predicate are [totientTail](#) and [certifiedKill](#).

Step 2: collapse the two parameters to one. Every multiple of a period is a period, so $\text{lcm}(1, \dots, t)$ is the universal-period envelope: any eventual period of a rational S divides it once t is large. Standing on that one diagonal ray covers all possible periods at once, and the position parameter drops out too.

Result 9 (diagonal reduction). *Write $\text{lcm}(1..t)$ for $\text{lcm}(1, \dots, t)$. If for every t_0 there exist $t \geq t_0$ and L with $\text{certifiedKill}(\text{lcm}(1..t), \text{lcm}(1..t), L)$, then S is irrational.*

Formalized: [LcmDiagonalReduction.lean:92](#) (`irrational_totient_series_of_lcm_diagonal_certificate_supply`).

Step 3: the cone, flatness, and completeness. On the cone $\{k \cdot \text{lcm}(1..t)\}$, rationality forces not one pairwise relation but a single fractional constant across the whole cone.

Result 10 (cone flatness). *If S is rational, then the relevant tail values share one fractional part across the whole lcm cone.*

Formalized: [LcmConeFlatness.lean:90](#) (`rational_totient_series_forces_lcm_cone_flatness`).

The certificates are exact, not a lossy heuristic: a certificate exists exactly when the object it certifies is non-integral. That turns the finite computations from experiments into receipts.

Result 11 (certificate completeness). *For all h, N ,*

$$(\exists L, \text{certifiedKill}(h, N, L)) \iff R_{N+h} - R_N \notin \mathbb{Z}.$$

Formalized: [LcmConeFlatness.lean:316](#) (`exists_certifiedKill_iff_tail_diff_notMem_int`). Finite certificates are receipts; non-integrality is the object. A joint refuter can interrogate a menu of cone vertices at once and is sharper than any pairwise test: it refutes the single-fractional-part model that rationality predicts on a finite sample of cone vertices ([LcmConeNonflat.lean:126](#) (`exists_nonintegral_pair_of_coneNonflatCert`)).

The certificates are non-empty. The initial segment of the required supply is machine-checked, so the reduction is not vacuous. By completeness, each of these is a kernel-checked proof of a specific non-integral tail difference, not a numerical experiment.

Result 12 (finite deposits). *The following are checked by the Lean kernel:*

- *`certifiedKill`($h, 12, 16$) for every period $h \in [1, 8]$;*
- *`certified kills` for every period $h \in [1, 16]$ at a fixed window;*
- *diagonal certificates `certifiedKill`($\text{lcm}(1..t), \text{lcm}(1..t), L$) for $t \in [1, 6]$, and at $t = 7, 8$;*
- *a tabulated joint cone-vertex certificate.*

Formalized: [certifiedKill_all_small](#), [certifiedKill_all_upto_sixteen](#), [certifiedKill_periodLcm_diagonal_upto_six](#) (with [certifiedKill_periodLcm_diagonal_at_seven](#), [certifiedKill_periodLcm_diagonal_at_eight](#)), and [coneNonflatCert_cells](#). These deposits show that each layer of the reduction is non-vacuous and, by completeness, exact. They are not evidence that the unbounded supply exists.

What is open. The reduction converts #249 into the statement that the certificate supply is *unbounded*: certificates exist for arbitrarily large parameters. By completeness (Result 11) this is equivalent to the tail differences being non-integral infinitely often, and equivalently to the Farey gap denominators of Section 6.1 being unbounded: the same obstruction seen as a rational denominator, an eventual binary period, and a flat tail on the lcm cone. That unboundedness is the remaining analytic question, and it is left open. The reduction is real; the closure is not asserted.

7 Rationality as a carry system (further infrastructure)

Four further modules extend the development sideways. They prove no new irrationality. Instead they pin down, in machine-checked form, what a rational value of the series in question would have to look like: as an integer carry orbit, as a Möbius-inverted support coordinate, as a residue-wrap cycle modulo the odd part of the denominator, and as a point of a greedy subset-sum geometry. Everything in this section is an equivalence or an unconditional constraint on

the rational side; nothing excludes a rational value of either open series, and each module states that boundary in its header. One hedge is repeated from the sources: the literature has not been audited for prior statements of these lemmas, so they are offered as formal infrastructure, not as claimed novelties.

7.1 Tail-orbit rigidity: a rationality criterion

For a coefficient sequence $c : \mathbb{N} \rightarrow \mathbb{N}$ with $c(n) \leq n$, write

$$X_c = \sum_{n \geq 1} \frac{c(n)}{2^n}, \quad T_c(N) = \sum_{j \geq 1} \frac{c(N+j)}{2^j},$$

the series and its scaled tail after N binary digits. Call an integer sequence u a *tempered orbit* for c with multiplier $v \geq 1$ if

$$u(N+1) = 2u(N) - vc(N+1) \text{ for all } N, \quad \text{and} \quad u(N)/2^N \rightarrow 0.$$

The recurrence is exact binary long division against the coefficient stream; the boundary condition forbids the homogeneous solution.

Result 13 (tail-orbit rigidity). *Let $c(n) \leq n$ for all n . Every tempered orbit is the scaled tail, $u(N) = vT_c(N)$ for all N ; and X_c is rational if and only if a tempered orbit exists for some multiplier $v \geq 1$.*

Formalized: [GenericTailOrbitRigidity.lean:171](#) (`temperedBinaryOrbit_eq_scaledTail`) (rigidity) and [GenericTailOrbitRigidity.lean:259](#) (`binaryCoeffSeries_rational_iff_exists_temperedBinaryOrbit`) (the criterion), restated in Mathlib’s irrationality vocabulary at [not_irrational_binaryCoeffSeries_iff_exists_temperedBinaryOrbit](#). The engine is one observation: a real orbit with $d(N+1) = 2d(N)$ and $d(N) = o(2^N)$ vanishes identically ([doublingOrbit_eq_zero_of_tempered](#)).

The tempered boundary is essential, and the source is deliberate about it: the homogeneous parasite $N \mapsto 2^N$ can be added to any orbit without disturbing the recurrence, so positivity of an orbit certifies nothing by itself, and positivity is nowhere advertised as an equivalent criterion. The linear-growth hypothesis covers both constants of Section 3: $\varphi(n) \leq n$ for the #249 coefficients, and $f_A(n) \leq \tau(n) \leq n$ for the #257 support coefficients $f_A(n) = \#\{a \in A : a \mid n\}$. The modules below instantiate the criterion on supports; the totient instantiation is not taken here.

7.2 Boolean–Möbius carry coordinates (the #257 direction)

At base 2 the support series of Section 5 is itself a binary coefficient series, $\sum_{n \in A} 1/(2^n - 1) = X_{f_A}$ ([erdosSupportSeries_two_eq_binaryCoeffSeries](#)). Two exact coordinate changes then turn a hypothetical rational value into a support-free object.

Result 14 (Boolean Möbius inversion, both directions). *On positive integers, $f_A = \mathbf{1}_A * \zeta$ and $\mu * f_A = \mathbf{1}_A$, where $\mathbf{1}_A$ is the indicator of A . Conversely, every integer-valued arithmetic function f whose Möbius transform is Boolean, $(\mu * f)(n) \in \{0, 1\}$ for all $n \geq 1$, satisfies $f = f_B$ for the support $B = \{n \geq 1 : (\mu * f)(n) = 1\}$ selected by that transform.*

Formalized: [BooleanMobiusCarry.lean:88](#) (`positiveSupportBitAF_mul_zeta`), [BooleanMobiusCarry.lean:106](#) (`moebius_mul_supportCoeffAF`), and, with no search or finiteness hypothesis, the converse [BooleanMobiusCarry.lean:192](#) (`eq_supportCoeffAF_booleanMobiusSupport_of_boolean`); the selected support is [booleanMobiusSupport](#).

On the carry side, Result 13 applies at $c = f_A$: the support series is rational exactly when a tempered carry orbit for f_A exists ([erdosSupportSeries_rational_iff_exists_tempered](#)

`Carry`); a displayed fraction p/q corresponds to an orbit U with multiplier q and initial value $U(0) = p$ (`support_fraction_iff_exists_temperedCarry`); and exact division recovers the coefficient from the orbit, $(2U(N) - U(N+1))/q = f_A(N+1)$ (`temperedCarry_quotient_eq_supportCoeff`). The elementary divisor-pair bound $\tau(n) \leq 2\lfloor\sqrt{n}\rfloor$ (`card_divisors_le_two_mul_sqrt`) confines the tails to a square-root strip, $T_{f_A}(N) \leq 2\sqrt{N} + 4$ (`binaryCoeffTail_supportCoeff_le_two_sqrt_add_four`), hence every orbit state to $0 < U(N) \leq q(2\sqrt{N} + 4)$ (`temperedCarry_pos_of_exists_pos_mem`, `temperedCarry_le_denominator_mul_two_sqrt_add_four`); and the strip is strong enough to re-derive the tempered boundary, so the analytic side condition can be traded for one inequality. The two coordinate changes compose.

Result 15 (certificate-level equivalence). *Fix an integer p and an integer $q \geq 1$. There is a nonempty support A of positive integers with $\sum_{n \in A} 1/(2^n - 1) = p/q$ if and only if there is an integer sequence U with $U(0) = p$ such that: every $U(N)$ is positive; $U(N) \leq q(2\sqrt{N} + 4)$; q divides $2U(N) - U(N+1)$ for every N ; and the Möbius transform of the quotient sequence $(2U(N) - U(N+1))/q$ is Boolean. The support is not guessed: it is reconstructed from the certificate by Result 14.*

Formalized: `BooleanMöbiusCarry.lean:776` (`exists_normalized_support_fraction_iff_exists_booleanMöbiusCarry`), with the certificate packaged as `BooleanMöbiusCarryCertificate` and the quotient normalized at index zero (`carryQuotient`). A worked fixture keeps the coordinates honest: the support $\{2, 3\}$ has value $\frac{1}{3} + \frac{1}{7} = \frac{10}{21}$ (`erdosSupportSeries_support23_eq_ten_div_twenty_one`), its orbit is the pure period-six cycle 10, 20, 19, 17, 13, 26 with multiplier 21 (`carryOrbit23_isTempered`), and the Möbius transform of the carry quotient recovers exactly $\{2, 3\}$ (`möbius_carryOrbit23_recovers_support`).

These are coordinates, not by themselves a strategy: nothing here excludes infinite Boolean carry paths, and no finite search is promoted to a completeness argument.

7.3 Residue wraps, reciprocal mass, and unbounded carries

Suppose now the support series equals $p/(2^c v)$ with v odd and A containing a positive element. Rationality makes the scaled tails integers: $u(N) = v T_{f_A}(c + N)$ is a positive integer, satisfies the carry recurrence $u(N+1) + v f_A(c + N+1) = 2u(N)$, and reduces modulo v to the doubling residues $r_N = p 2^N \bmod v$ (`exists_shifted_odd_tail_nat_states_of_support_fraction`). Doubling a least residue either stays under v or wraps past it exactly once, so each step emits a wrap digit $k_N = \lfloor 2r_N/v \rfloor \in \{0, 1\}$, and over any cycle length h with $2^h \equiv 1 \pmod{v}$ the repetend identity holds:

$$\sum_{N < h} r_N = v \cdot w, \quad w = \sum_{N < h} k_N,$$

with $w \geq 1$ when p is coprime to $v > 1$. Formalized: `RationalSupportCarrySkeleton.lean:195` (`sum_doublingResidue_eq_mul_wrapCount`) and `doublingWrapCount_pos`; residues, digits, and counts are `doublingResidue`, `doublingWrapDigit`, `doublingWrapCount`.

Result 16 (one-wrap classification). *Let $v > 1$ be odd, p coprime to v , and $h \geq 1$ with $2^h \equiv 1 \pmod{v}$. If the cycle of p wraps exactly once, then $v = 2^h - 1$ and the starting residue is a power of two: $r_0 = 2^a$ for some $a < h$.*

The one-wrap cycles are thus exactly the Mersenne repetends $2^a/(2^h - 1)$. Formalized: `RationalSupportCarrySkeleton.lean:440` (`one_doublingWrap_classification`), from the generic closed-cycle form `one_wrap_cycle_classification`, and instantiated at the true order $h = \text{ord}_v(2)$ (`one_oddOrder_doublingWrap_classification`, with `oddDoublingOrder`). The classification is algebraic; a finite check of 446 coprime starting residues across twelve modulus/order rows is retained as kernel-checked validation, not as the proof (`orderWrapValidatedStartCount_eq_446`, `orderWrapValidation446_passes`, `orderWrap_minima_table_passes`).

The analytic bridge is Cesàro averaging. Write $\rho(A) = \sum_{a \in A} 1/a$ for the reciprocal mass ([reciprocalMass](#)). When the reciprocal family is summable, the mean of $f_A(1), \dots, f_A(N)$ converges to $\rho(A)$ — the density of the multiples of a , summed over the support — and the mean of the scaled tails has the same limit ([RationalSupportCarrySkeleton.lean:971](#) ([tends to_supportCoeff_mean_reciprocalMass](#)), [tendsto_binaryCoeffTail_supportCoeff_mean_reciprocalMass](#)). The divergent case is carried as an explicit disjunction, not through the convention that a divergent `tsum` is zero ([ReciprocalMassDivergesOrAtLeast](#)). Each tail state splits exactly as $u(N) = v e_N + r_N$ with integral excess $e_N = \lfloor u(N)/v \rfloor \geq 0$ ([oddTailExcess](#)).

Result 17 (order-wrap lower bound, with exact excess mean). *Let A and $p/(2^c v)$ be as above with $v > 1$ odd, and let w be the wrap count of p over one cycle of length $h = \text{ord}_v(2)$. Then $\sum_{a \in A} 1/a$ diverges or $\rho(A) \geq w/h$; if moreover p is coprime to v , then $w \geq 1$, so $\rho(A) \geq 1/\text{ord}_v(2)$. In the summable case the bound is the periodic part of an exact identity: the Cesàro mean of the excess e_N converges automatically, and*

$$\rho(A) = \frac{w}{h} + \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n < N} e_n.$$

Formalized: [RationalSupportCarrySkeleton.lean:1479](#) ([support_fraction_oddOrder_wrapRatio_le_reciprocalMass](#)), [RationalSupportCarrySkeleton.lean:1503](#) ([one_div_oddOrder_le_reciprocalMass_of_support_fraction](#)), the divergence-honest form [RationalSupportCarrySkeleton.lean:1542](#) ([support_fraction_one_div_oddOrder_reciprocalMassDivergesOrAtLeast](#)), and the excess mean [RationalSupportCarrySkeleton.lean:1359](#) ([tendsto_oddTailExcess_mean](#)) (fraction-facing shifted form [exists_shifted_oddTailExcess_mean_of_support_fraction](#), named-limit form [reciprocalMass_eq_wrapRatio_add_oddTailExcessMean](#)). A rational value thus ties the odd part of its denominator to the sparsity of its support: a summable support with small $\rho(A)$ can only display denominators whose odd part has small multiplicative order of 2.

Result 18 (collision strengthening at common multiples). *In the summable case, let $F \subseteq A$ be finite and let L be a common multiple of F . Then*

$$\rho(A) \geq \frac{w}{h} + \frac{\lceil (|F| - 1)/2 \rceil}{L}.$$

In particular, an infinite support whose value is a dyadic rational $p/2^c$ has $\sum_{a \in A} 1/a$ divergent or $\rho(A) > 1$.

The mechanism is a collision: at every positive multiple of L at least $|F|$ support divisors land on the same index, so $f_A \geq |F|$ there, while the exact carry equation $f_A(n) + e_n = k_{n-1} + 2e_{n-1}$ with $k_{n-1} \leq 1$ lets one step absorb at most one unit — so the preceding excess must spike to at least $\lceil (|F| - 1)/2 \rceil$, and the spikes recur with density $1/L$. Formalized: [RationalSupportCarrySkeleton.lean:2062](#) ([booleanCollision_wrap_bound_of_common_multiple](#)) (denominator-shifted form [RationalSupportCarrySkeleton.lean:1988](#) ([shiftedBooleanCollision_wrap_bound_of_common_multiple](#))), the carry equation [supportCoeff_add_oddTailExcess_succ_eq_wrap_add_two_excess](#), and the dyadic corollary [RationalSupportCarrySkeleton.lean:2225](#) ([dyadic_support_fraction_reciprocalMass_diverges_or_gt_one](#)) (via [RationalSupportCarrySkeleton.lean:2141](#) ([one_lt_reciprocalMass_of_dyadic_support_fraction_of_two_pos_mem](#))).

Result 19 (carry states are unbounded over infinite supports). *Let A be infinite with value $p/(2^c v)$, $v \geq 1$. Then the positive integer carry state $u(N) = v T_{f_A}(c + N)$ is unbounded: for every B there is an N with $u(N) > B$.*

The proof picks $2B + 1$ positive support elements; at a common multiple L of all of them, the local inequality $1 + v|F| \leq 2u(L - c - 1)$ pushes the state past B . Formalized: [RationalSupportCarrySkeleton.lean:2402](#) (`exists_unbounded_shifted_odd_tail_nat_state_of_support_fraction`), from [shifted_state_unbounded_of_infinite_support](#) and the local inequality [one_add_mul_card_le_two_mul_shifted_state](#); the $\{2, 3\}$ fixture revalidates it at sixty-six common multiples ([carryOrbit23_common_multiple_bound_validation66](#)). So a rational value over an infinite support cannot run on a bounded carry alphabet. That closes finite-state readings of the carry system; it does not close the problem. Nothing in Results 16–19 prevents an infinite support from satisfying every one of these constraints.

7.4 Greedy geometry of the #257 value set

The fourth module studies the set of all values a base-2 support series can take. For $n \geq 1$ put $w_n = 1/(2^n - 1)$, let $T(n)$ be the mass strictly after exponent n , and let

$$\mathcal{A} = \left\{ \sum_{n \in A} w_n : A \subseteq \mathbb{N}, 0 \notin A \right\},$$

the *Mersenne achievement set*, the exponent 0 being normalized away as analytically invisible. Coding by supports of positive exponents is literally the kernel’s support series ([positiveMersenneSupportValue_eq_erdosSupportSeries](#); the set is [mersenneAchievementSet](#)). In this language the base-2 #257 statement reads: the rational members of $\mathcal{A}$ are exactly the finite subset sums. Nothing below decides that.

Result 20 (superincreasing geometry and greedy membership). *For every $n \geq 1$ the weight dominates the whole tail after it, $T(n) < w_n$, with the two-scale gap $w_n - T(n) = \frac{2}{3}4^{-n} + O(8^{-n})$ and explicit valid O -constant 3. Consequently a real x belongs to $\mathcal{A}$ exactly when $x \geq 0$ and every greedy residual of x is at most the remaining tail; and normalized support coding is injective, so each member of $\mathcal{A}$ has a unique support, which the greedy recursion recovers bit by bit.*

Formalized: [GreedyAchievementSet.lean:183](#) (`mersenneTail_lt_weight`), [GreedyAchievementSet.lean:422](#) (`mersenneGap_isBig0With`) (explicit bound [mersenneGap_asymptotic_bound](#)), [GreedyAchievementSet.lean:750](#) (`mem_mersenneAchievementSet_iff_greedy_survival`), and [GreedyAchievementSet.lean:759](#) (`positiveMersenneSupportValue_injective_normalized`), transferred to the kernel series at [erdosSupportSeries_two_injective_normaliz](#)[ed](#). The nonnegativity guard in the membership criterion is necessary: without it a negative target would survive every level while lying outside $\mathcal{A}$.

Result 21 (exact rational runs and death certificates). *The greedy recursion executed in exact rational arithmetic agrees step by step with the real recursion under casting. A finite decidable certificate — the exact rational residual at some level is at least an exact rational upper enclosure of the remaining tail — soundly proves non-membership in $\mathcal{A}$; the level-one instance shows $3/4 \notin \mathcal{A}$. And a rational number already known to lie in $\mathcal{A}$ has computable support bits.*

Formalized: [GreedyAchievementSet.lean:574](#) (`cast_greedyMersenneRemainderRat`); [GreedyAchievementSet.lean:827](#) (`certifiedGreedyMersenneDeath_not_mem`), with the certificate [CertifiedGreedyMersenneDeath](#) and the enclosure [mersenneTailUpperRat](#); [GreedyAchievementSet.lean:849](#) (`three_fourths_not_mem_mersenneAchievementSet`); and [GreedyAchievementSet.lean:933](#) (`rational_member_support_bit_iff`). The certificates are one-sided: survival through any finite depth proves nothing about membership, and no decidability of membership in $\mathcal{A}$ is asserted. The module also declines to formalize the topological or measure-theoretic geometry of $\mathcal{A}$; in particular the phrase “fat Cantor set” is not promoted to a Lean-checked theorem.

What this section does not do. No statement above excludes a rational value of the totient constant or of any infinite-support series. The four modules supply coordinates (Results 13, 14, 15), constraints (Results 16–19), and a decision geometry for the value set (Results 20, 21). The closing argument — excluding every infinite Boolean carry path, or exhibiting an unbounded certificate supply — is exactly what Sections 5 and 6 leave open.

8 Where this sits: the Erdős landscape

The table records how the development relates to the numbered problems it is near. Statuses are as listed on the individual problem pages erdosproblems.com/{249, 257, 1049, 69, 250, 258} in July 2026; none of the external results is re-proved here.²

problem	status	relation in this repository
#249 $\sum \varphi(n)/2^n$	open	the target; unconditional denominator exclusion and a conditional reduction, no solution
#257 $\sum_{n \in A} 1/(2^n - 1)$	open	named infinite-support cases and the full-support ($A = \mathbb{N}$) case, not the universal statement
#1049 $\sum 1/(t^n - 1)$, $t \in \mathbb{Q}_{>1}$	open (rational t)	the integer-base case $b \geq 2$ is formalized (Result 4); the rational case is untouched
#69 $\sum \omega(n)/2^n$	solved [7]	the prime-support case of #257; only the identity bridge to $\sum_p 1/(2^p - 1)$ is formalized, not the irrationality
#250 $\sum \sigma(n)/2^n$	solved [4]	the ladder neighbour $L(\text{Id})$; cited, not re-proved
#258 $\sum \tau(n)/(a_1 \cdots a_n)$	solved	not addressed here; the monotone φ/σ sequel is the sibling territory of #249

In short: this is a formalization of Erdős–Borwein-type irrationality together with an unconditional record and a finite-certificate reduction for the open totient constant. It is not a solution to #249 or to the universal #257.

9 What is proved, cited, and open

Proved (machine-checked, no sorry, no admit). Irrationality of $\sum 1/(b^n - 1)$ for every $b \geq 2$ and of the named support cases (Results 4, 5); the ladder identities and the Möbius-square and coprimality reformulations (Results 2, 3, 7); the unconditional denominator exclusion to 7.96×10^{34} (Result 6); the full chain of conditional #249 reductions with certificate completeness and a non-empty initial supply (Results 8–12); and the carry-coordinate equivalences, rational-value constraints, and greedy value-set geometry of Section 7 (Results 13–21).

Cited, not formalized here. Transcendence of $\sum \sigma(m)/2^m$ (Nesterenko 1996 [4]) and the q -zeta anchor (Postelmans–Van Assche [5]). They are named as ladder neighbours.

Open, and not solved here. Irrationality of S (#249); the universal #257 statement. The conditional reductions turn #249 into “the certificate supply is unbounded”, which is the remaining analytic question.

²The irrationality of S (#249) and the universal #257 statement are open. #1049 is open for rational $t > 1$, with the integer-base case attributed to Erdős. #69 (Tao–Teräväinen [7]), #250 (Nesterenko 1996 [4]), and #258 (recorded as proved in Lean) are solved; none is re-proved here.

The repository states these boundaries in code as machine-checked non-claims: `not_erdos_249_solution`, `not_erdos_257_solution`, and four provenance non-claims ([NON_CLAIMS.md](#)).

10 Verification and reproducibility

The package is checked by the Lean 4 kernel against Mathlib, pinned in [lake-manifest.json](#) at toolchain `leanprover/lean4:v4.29.1`. Three facts matter for trust.

- There is no `sorry` and no `admit` anywhere in the package.
- Every finite computation is discharged by `decide`, checked by the kernel. There is no use of `native_decide`, so no compiler is trusted.
- The package has more than 1,600 theorem declarations. The assembled microkernel is [CertificateKernel.lean](#) (about 18,900 lines); the machine-generated finite certificates are [GeneratedCertificates.lean](#) (about 27,700 lines) plus per-base shards, each instance closed by `decide`.

To build and check:

```
lake exe cache get      # fetch the prebuilt Mathlib cache
lake build              # build and kernel-check the package
```

Trust rests on `lake build` alone: it elaborates every declaration and the Lean 4 kernel checks each proof term, including every `decide`, against Mathlib at the pinned toolchain. There is no separate trusted step and no compiler in the loop. Continuous integration runs the same build on every push.

11 A sibling project, in progress

Separately from this repository, I am working on the [Ramanujan Machine Challenge](#) (ten problems, released July 2026, on proofs or symbolic derivations of explicit formulas for constants such as π , e , Catalan’s constant, and zeta values). That work is a different Lean lane and is not part of this snapshot: no challenge target is claimed solved or formalized here, and nothing in this paper depends on it. A coarse status map for that project lives in the repository README.

A Index of principal declarations

Each row links to the declaration at commit [de7a360](#).

statement	source line (and declaration)
$\sum 1/(b^n - 1)$ irrational, all $b \geq 2$	CertificateKernel.lean:7999 irrational_erdosSum_full_support
Erdős–Borwein E irrational	CertificateKernel.lean:8006 irrational_erdosBorwein_series
factorial support	CertificateKernel.lean:5706 irrational_erdosSum_factorial_support
power-of-two support	CertificateKernel.lean:5730 irrational_erdosSum_two_pow_support
pairwise-coprime supports	CertificateKernel.lean:10447 irrational_erdosSupportSeries_pairwise_coprime
$L(\mu) = 1/2$	MersenneLambertLadder.lean:527 tsum_moebius_div_two_pow_sub_one_eq_half
$L(\varphi) = 2$	MersenneLambertLadder.lean:515 tsum_totient_div_two_pow_sub_one_eq_two
positive lift $L(A) = S$	CertificateKernel.lean:18116 tsum_primWeight_div_two_pow_sub_one_eq_totient_series
Möbius-square lens	CertificateKernel.lean:18125 totient_series_eq_half_add_moebius_mersenne_square
$S = \frac{1}{2} + \Pr(\gcd = 1)$	CertificateKernel.lean:18228 totient_series_eq_half_add_visible_coprime_pairs
$S \neq p/q$ for $q \leq 7.96 \times 10^{34}$	CertificateKernel.lean:18055 tsum_totient_div_pow_two_ne_ratCast_of_den_le_79639646646701375323355774875831053
tail-period reduction	TotientTailPeriodKiller.lean:386 irrational_totient_series_of_certificate_supply
diagonal reduction	LcmDiagonalReduction.lean:92 irrational_totient_series_of_lcm_diagonal_certificate_supply
cone flatness from rationality	LcmConeFlatness.lean:90 rational_totient_series_forces_lcm_cone_flatness
certificate completeness	LcmConeFlatness.lean:316 exists_certifiedKill_iff_tail_diff_notMem_int
joint cone refuter	LcmConeNonflat.lean:126 exists_nonintegral_pair_of_coneNonflatCert
non-empty certified-kill segment	TotientTailPeriodKiller.lean:396 certifiedKill_all_small
tail-orbit rationality criterion	GenericTailOrbitRigidity.lean:259 binaryCoeffSeries_rational_iff_exists_temperedBinaryOrbit
Boolean carry certificate $\Leftrightarrow$ support fraction	BooleanMobiusCarry.lean:776 exists_normalized_support_fraction_iff_exists_booleanMobiusCarry
one-wrap classification	RationalSupportCarrySkeleton.lean:440 one_doublingWrap_classification
order-wrap mass bound	RationalSupportCarrySkeleton.lean:1503 one_div_oddOrder_le_reciprocalMass_of_support_fraction
unbounded carry states	RationalSupportCarrySkeleton.lean:2402 exists_unbounded_shifted_odd_tail_nat_state_of_support_fraction
greedy membership criterion	GreedyAchievementSet.lean:750 mem_mersenneAchievementSet_iff_greedy_survival

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- [7] T. Tao and J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, arXiv:2512.01739 (2025); settles Erdős #69, the irrationality of $\sum_n \omega(n)/2^n = \sum_p 1/(2^p - 1)$.
- [8] T. Bloom (ed.), *Erdős Problems*, erdosproblems.com, accessed July 2026.
- [9] The Ramanujan Machine, *The Ramanujan Challenge* (2026), ramanujanmachine.com/ramanujan-challenge.